Topic 1.9 - Rational Functions and Vertical Asymptotes

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: ______________________ Class: ______________ Date: ______________

Why this topic matters

A vertical asymptote describes unbounded output near a finite excluded input. Identifying the asymptote is only the first step: strong analysis explains each one-sided limit by simplifying, checking multiplicity, and tracking signs rather than relying on a sketch.

Learning targets

- Identify vertical asymptotes after cancellation and distinguish them from holes.
- Determine one-sided infinite behavior using sign and multiplicity.
- Construct and interpret rational functions with specified asymptotic behavior.

Language and structure

Term	Meaning in this topic
vertical asymptote	An excluded input near which output magnitude grows without bound.
one-sided limit	Behavior as x approaches from values less than or greater than the input.
even denominator multiplicity	The denominator factor keeps the same sign on both sides.
odd denominator multiplicity	The denominator factor changes sign across the asymptote.
divergence	Unbounded growth toward $+\infty$ or $-\infty$.

Launch - reason before calculating

Compare $R(x) = \frac{x+1}{x-2}$ and $S(x) = \frac{x-3}{(x+1)^2}$ near their vertical asymptotes. Why do the one-sided directions differ?

Concept 1: Simplify before identifying asymptotes

Core idea

Factor and cancel common factors while retaining restrictions. Uncanceled real denominator zeros are vertical asymptotes; canceled denominator zeros are holes. Positive quadratics may add no real asymptotes.

Worked example - annotate the reasoning

Problem. Classify excluded inputs of $F = \frac{(x-1)(x+2)}{(x-1)(x-4)}$.

Model reasoning. $x = 1$ is a hole after cancellation; $x = 4$ remains an uncanceled denominator zero and is a vertical asymptote.

Check your understanding

Independent | No calculator

Classify $G = \frac{x^2-9}{(x-3)^2(x+1)}$.

Concept 2: One-sided signs and multiplicity

Core idea

Near $x = a$, freeze factors that stay nonzero at their signs and focus on $(x-a)^m$. If m is odd, denominator sign changes; if even, it does not. The nearby numerator sign determines whether the unbounded values are positive or negative.

Worked example - annotate the reasoning

Problem. Find one-sided behavior of $R = \frac{x+1}{x-2}$ at 2.

Model reasoning. Numerator is near $3 > 0$. $x-2$ is negative left and positive right, so $R \to -\infty$ from the left and $+\infty$ from the right.

Check your understanding

Independent | No calculator

Find one-sided behavior of $S = \frac{x-3}{(x+1)^2}$ at -1.

Concept 3: Multiple asymptotes and construction

Core idea

Analyze each asymptote locally; other factors contribute fixed signs. To construct a requested pattern, choose denominator multiplicities and a numerator whose sign at each asymptote creates the required directions.

Worked example - annotate the reasoning

Problem. Analyze $T = \frac{x-1}{(x+2)^2(x-3)}$ at -2 and 3.

Model reasoning. Near -2, numerator and $x-3$ are both negative, and the square is positive, so both sides $+\infty$. Near 3, numerator and square are positive, while $x-3$ changes sign: left $-\infty$, right $+\infty$.

Check your understanding

Independent | No calculator

Construct a function with vertical asymptote 0 and both sides approaching $-\infty$.

Representation laboratory

Connect at least two representations

A graph has vertical asymptote $x = 1$ with left $+\infty$ and right $-\infty$, and asymptote $x = -3$ with both sides $+\infty$. Give one factored rational function that matches.

Challenge laboratory

Challenge - justify every claim

Construct a rational function with zero 1, vertical asymptote -2 approached by $+\infty$ from both sides, and vertical asymptote 3 approached by $-\infty$ from the left and $+\infty$ from the right.

Error clinic and calculator strategy

Common errors to diagnose

- Calling a canceled denominator zero a vertical asymptote.
- Stating $\pm\infty$ without separate one-sided analysis.
- Using denominator multiplicity alone while ignoring numerator and other-factor signs.
- Treating infinity as an output value attained by the function.

Technology decision

Use a table approaching from both sides with successively smaller distances and graph separate branches. Auto-scaling can hide direction or make a hole look like an asymptote. Exact sign analysis should lead; technology should confirm.

AP-style synthesis

Multi-part reasoning task

Let $R(x) = \frac{(x+4)(x-1)}{(x+2)^2(x-3)}$.

A. Identify zeros and vertical asymptotes.

B. Determine both one-sided limits at $x = -2$.

C. Determine both one-sided limits at $x = 3$.

Exit ticket

Before you leave

1. What distinguishes a hole from a vertical asymptote?
2. How does even denominator multiplicity affect side signs?
3. Why are one-sided limits necessary?